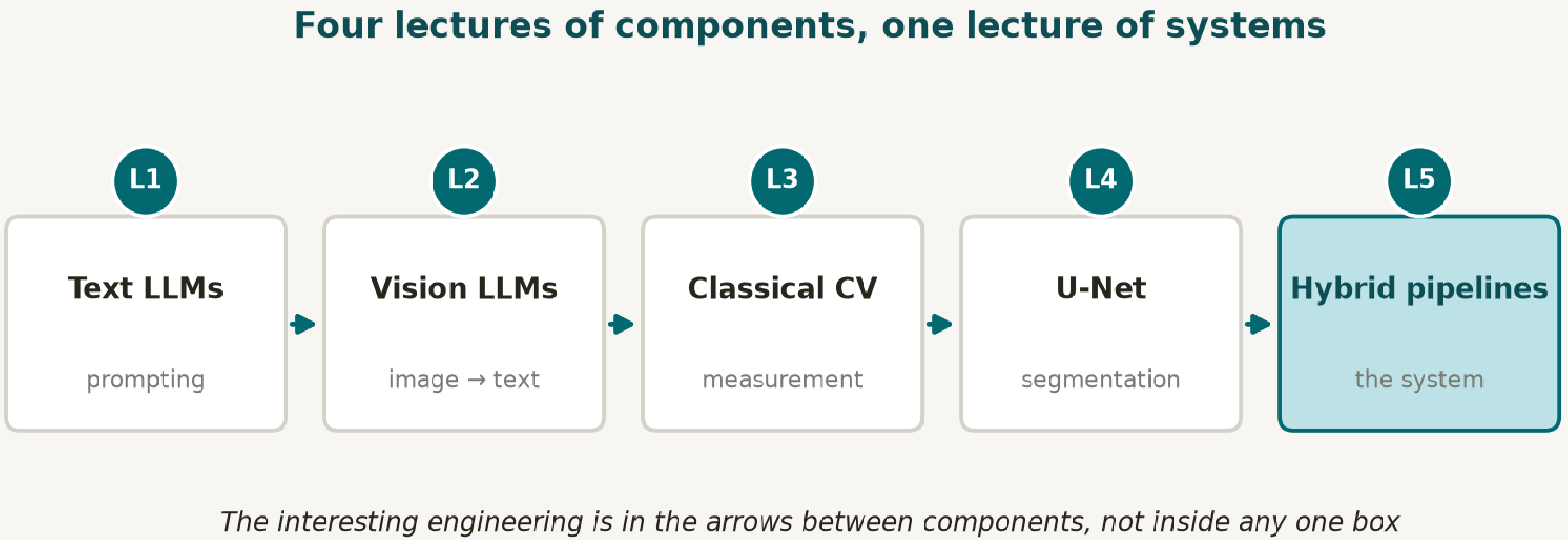
Hybrid Pipelines and Medical AI in Practice

How real medical AI systems are built, evaluated and deployed — and how they fail

Biomedical Image Analysis · Lecturer Mykola Korabel

Four lectures of components, one lecture of systems

You already know how to build the pieces. Today we connect them — and ask where systems break.



Sources: [Course lecture plan]

What you should be able to do by the end

Five outcomes. The first three are examinable; the last two are the point.

1. Describe any medical AI system as a pipeline of components with a structured intermediate representation.
2. Walk a fault tree across that pipeline: for each arrow, name what can go wrong and how it would be caught.
3. Apply two design principles — maximise the auditable component; forbid the narrative from inventing facts.
4. Distinguish research use from clinical use, and say what regulatory clearance does and does not mean.
5. Judge which part of a system you would trust least — and defend that judgement with reasons.

The central claim

Real medical AI systems are almost never a single model.

They are pipelines, in which each component does what it is best at, and in which a structured intermediate representation — typically a JSON (JSON - JavaScript Object Notation - a simple text format used to store data, send data over the internet, and connect different software programs) record or small DataFrame — is the source of truth that everything downstream, including the human reviewer, can audit.

The seductive picture

- An image goes in one end of a large model; a clinical sentence comes out the other.
- **Nothing to inspect in between.**
  - Performs well on benchmarks.
  - Cannot be tested, explained, monitored or cleared.

The picture that actually ships

- Half a dozen small components, a table of numbers in the middle, a named person who signs the output.
- **Every number is auditable.**
  - Drift is visible because numbers are logged.
  - A regulator can ask what it measures and how accurately.

Why “one big model” is the wrong mental model

Not because it performs badly — sometimes it performs very well — but because of what you give up.

You cannot test it

No intermediate quantity to compare against ground truth. You can only score the final sentence — and sentences are hard to score.

You cannot explain it

There is no artefact between the pixels and the claim. “The model said so” is the whole of the explanation.

You cannot monitor it

Nothing numeric comes out, so nothing can be tracked over time. Drift stays invisible until somebody complains.

You cannot repair it

A failure has no location. Retraining the whole model is your only lever, and it may break what already worked.

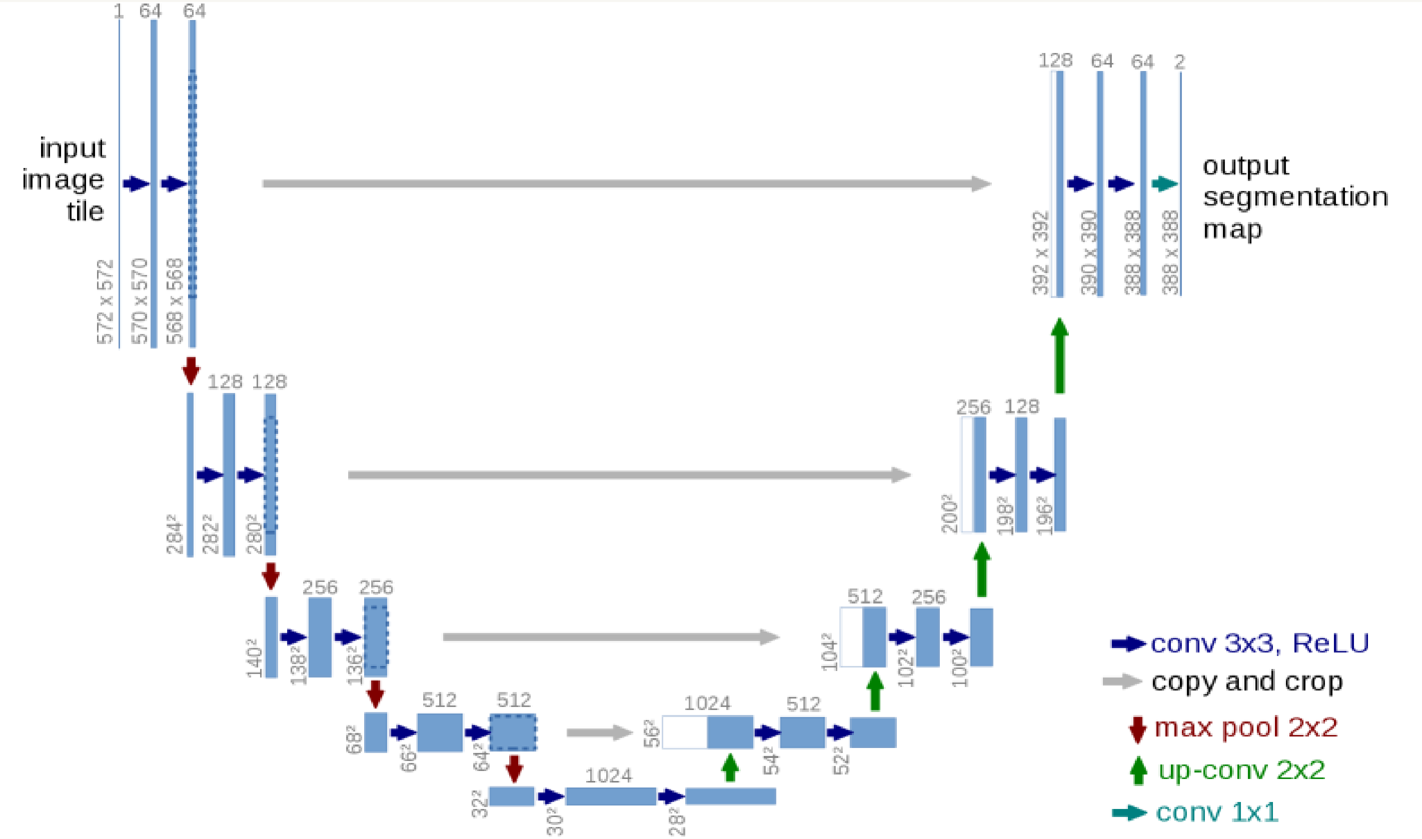
You cannot get it cleared

Regulators ask what the device measures and how accurately. “It describes the image” is not an answer.

You cannot assign responsibility

When the output is wrong, no component owns the error — a legal problem long before it is a technical one.

U-Net: one component, not a system



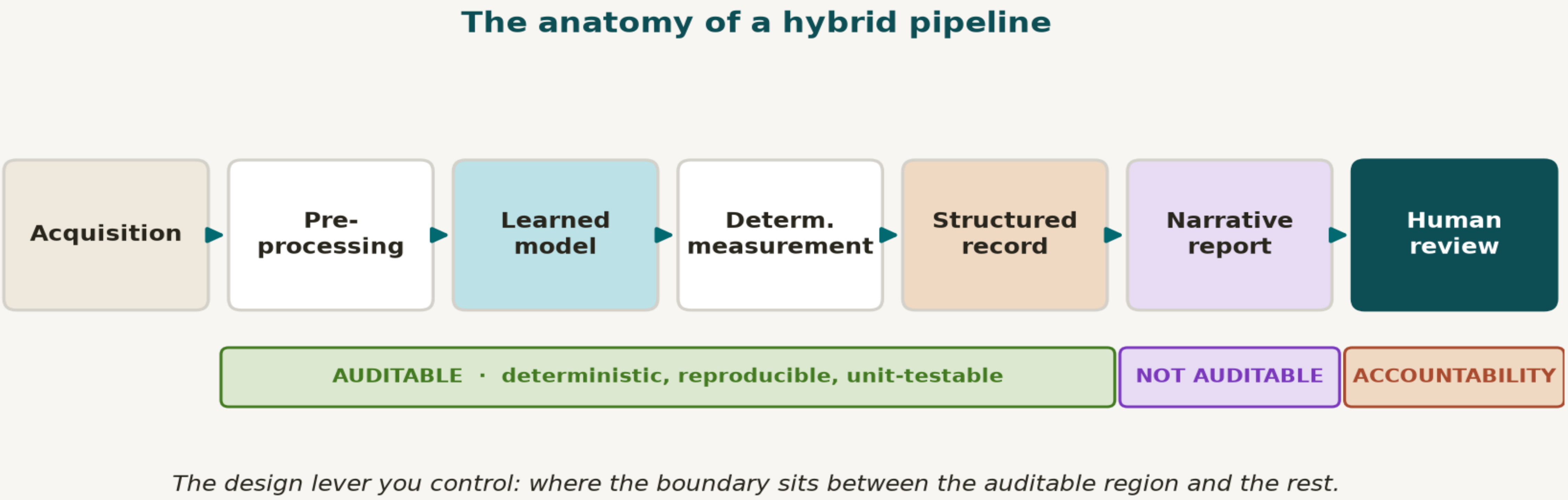
U-Net architecture (Ronneberger et al., MICCAI 2015, Fig. 1)

What this figure teaches

- **U-Net solves exactly one box: learned segmentation.**
- It does not acquire images, measure objects, format records, or narrate findings.
  - Encoder–decoder + skip connections: deep semantics meet fine detail.
- **A great model. A terrible system, on its own.**
- Every deployed system wraps it in preprocessing, measurement, and a structured record.

Sources: Ronneberger et al., U-Net (2015)

The anatomy of a hybrid pipeline



- Seven roles: acquisition → preprocessing → learned model → deterministic measurement → structured record → narrative → human review. The lever you control is where the auditable boundary sits.

Each component does what it is best at

The allocation of work is a design decision — and the one most often got wrong.

Task	Best tool	Why
Find where the object is	Learned model (U-Net)	Shape and texture are hard to hand-code; this is what deep learning is for
Measure how big it is	Deterministic code	Exact, reproducible, unit-aware, testable against a phantom
Decide whether it is unusual	Explicit rule / fitted model	A threshold you can read, argue with, and change without retraining
Describe it in English	Language model	Fluent, well-hedged prose is genuinely hard to template
Decide what to do about it	A named human clinician	Accountability cannot be delegated to something that cannot be held to account

Read the last row twice. It is the only row that is not a technical claim.

The structured record is the source of truth

A real structured record: the contract between perception and language

record.json · schema v2.3

```
{
  "schema_version": "2.3",
  "model_version": "unet-v4.1",
  "image_id": "slide_07_tile_22.tif",
  "objects": [
    { "id": 1, "label": "nucleus",
      "area_px2": 842, "perimeter_px": 108.3,
      "eccentricity": 0.71, "mean_intensity": 0.62 },
    { "id": 2, "label": "nucleus",
      "area_px2": 1210, "perimeter_px": 131.7,
      "eccentricity": 0.83, "mean_intensity": 0.58 }
  ],
  "provenance": { "mask_ref": "masks/s07_t22.nii.gz",
                  "code_ref": "git@abc1234" }
}
```

- **Units live in field names**
- **Provenance pins every number**
- **Versioned schema + model**
- **Reviewer can audit in seconds**

The dividing line: auditable vs unauditable

AUDITABLE

- **Preprocessing → measurement → structured record**
- Deterministic: run twice, get the same answer.
- A specific line of code owns each number.
- Unit-testable against phantoms with known answers.
  - JSON or DataFrame — the choice matters less than committing to one and writing the schema down.

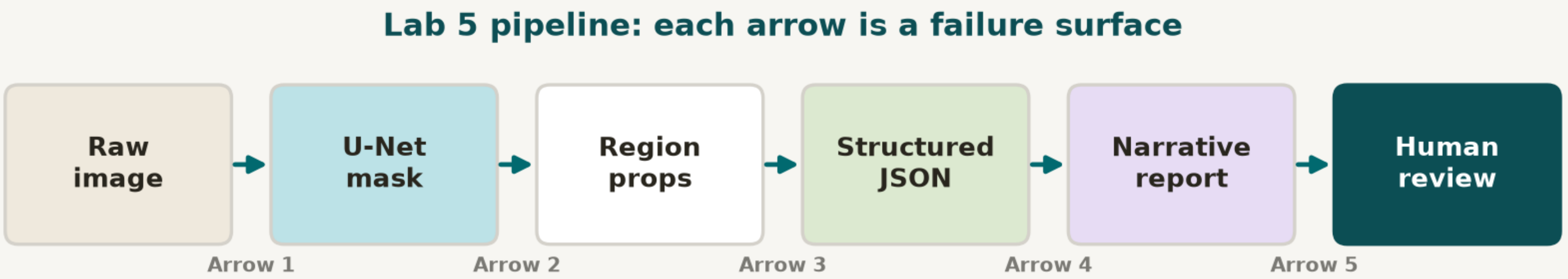
NOT AUDITABLE

- **Narrative report (LLM)**
- Fluent, useful, and fundamentally not reproducible.
- Run it twice, get two different sentences.

ACCOUNTABILITY

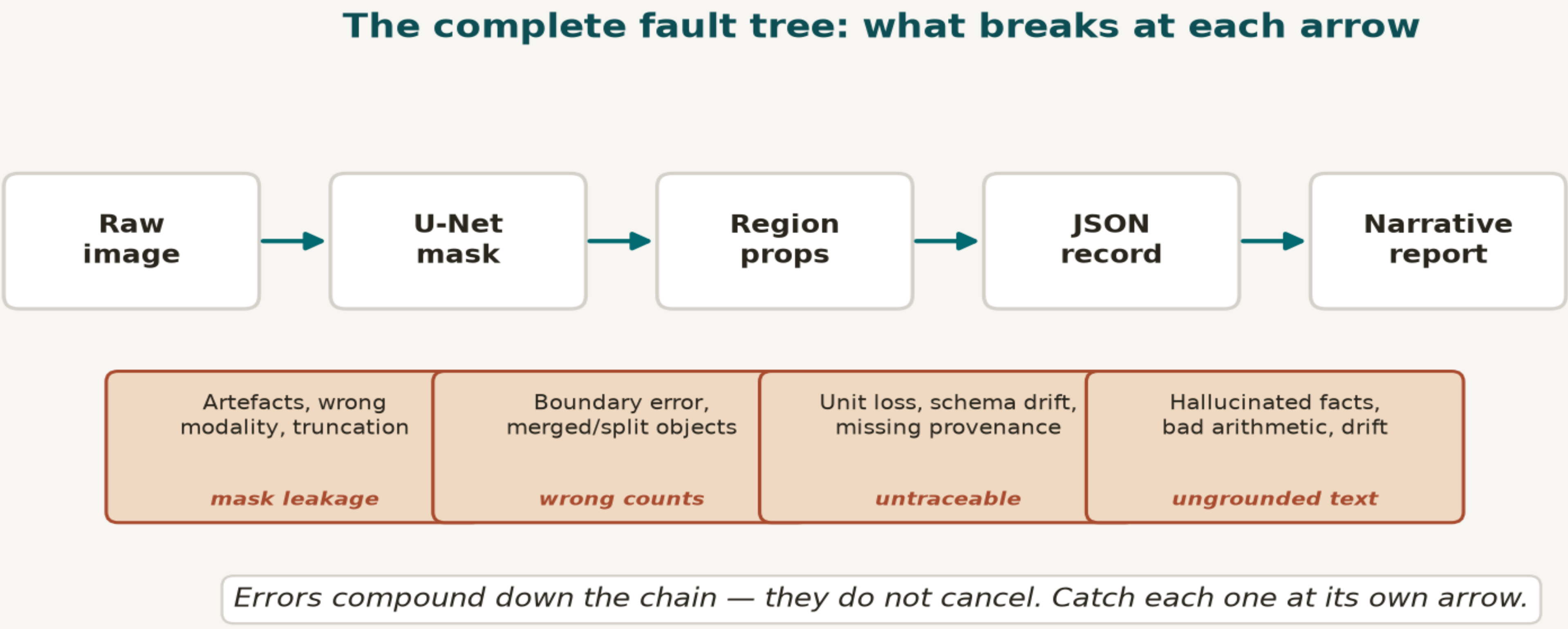
- **Human review**
- A named clinician signs the output.
- The DataFrame is the evidence; the JSON is the message; the signature is the accountability.

The Lab 5 pipeline, end to end



- Raw image → U-Net mask → region properties → structured JSON → narrative report → human review. Each arrow is a place where a small error can become a clinical mistake.

Every arrow is a failure surface



Arrow 1 — Raw image → U-Net mask

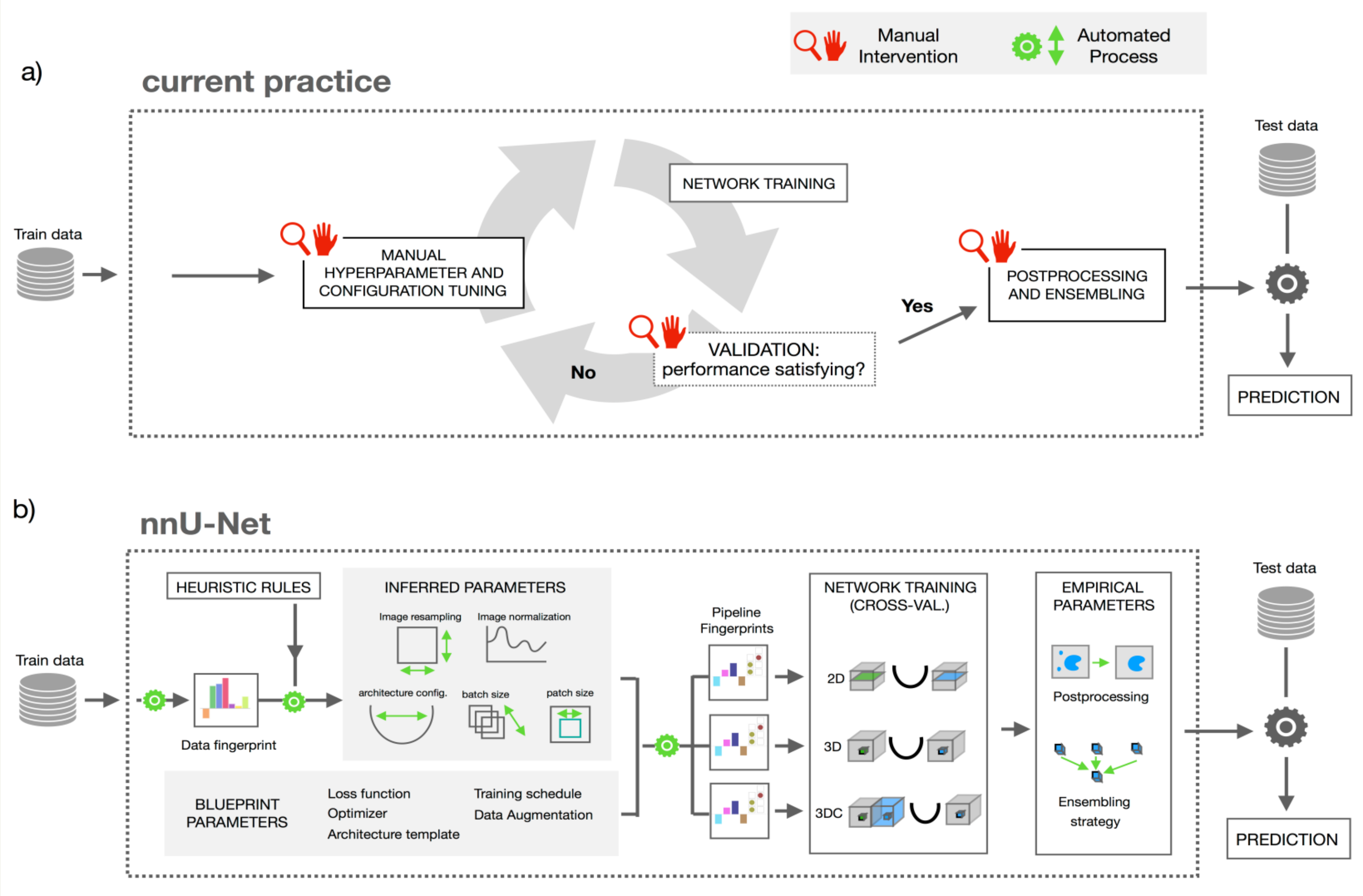
What goes wrong

- Artefacts, truncation, wrong modality fed in silently.
- Domain shift: model trained on scanner A, deployed on scanner B.
- Boundary leakage: two objects merge into one mask.
- Small lesions dropped below the model's confidence.
- **The mask "looks plausible" — failure is invisible to the eye.**

How you catch it

- Schema-validated input: modality, body-part, pixel spacing checked before inference.
- Hold-out test set per scanner/site; Dice + IoU per class.
- Sanity metrics on every mask: area, count, convexity — flag outliers.
- Uncertainty / abstention: low-confidence masks go to human queue.
  - Versioned model + versioned mask on disk (provenance).

Segmentation pipelines are more than an architecture



nnU-Net: self-configuring pipeline vs. manual practice (Isensee et al., Nat. Methods 2021, Fig. 2)

The lesson

- **The winning segmentation system is not a cleverer U-Net.**
- It is preprocessing, dataset fingerprinting, inferred hyperparameters, training, and postprocessing — all configured automatically.
- **“Pipeline, not model” is the empirical lesson of medical segmentation.**

Sources: Isensee et al., nnU-Net (2021)

Arrow 2 — U-Net mask → Region properties

What goes wrong

- A 5% boundary error becomes a 30% area error — area scales with radius².
- Merged/split objects give wrong counts and wrong per-region stats.
- Pixel units vs. physical units: forgetting spacing converts mm² to px².
- Holes and protrusions change eccentricity and solidity dramatically.
- **Measurement is deterministic — so the error is reproducible, and easy to miss.**

How you catch it

- Units in field names (area_mm2) so they cannot be silently lost.
- Cross-check: area vs. perimeter vs. convex area; flag impossible ratios.
- Phantoms with known geometry run nightly as a regression test.
    - Version the regionprops code (git ref in provenance).
- Compare counts to a recent rolling average; investigate spikes.

Arrow 3 — Region properties → Structured JSON

What goes wrong

- Schema drift: a new field appears, downstream code has never seen it.
- Silent type coercion: 842 becomes "842" becomes NaN downstream.
- Provenance dropped: a number with no code/model ref is untraceable.
- Row order mistaken for meaning; IDs not stable.

How you catch it

- Schema is versioned; consumer rejects unknown versions loudly.
- Strict typing + validation on write, not on read.
- Every field carries provenance: image_id, mask_ref, code_ref, model_version.
    - The record is immutable; corrections append, never overwrite.

Arrow 4 — Structured JSON → Narrative report

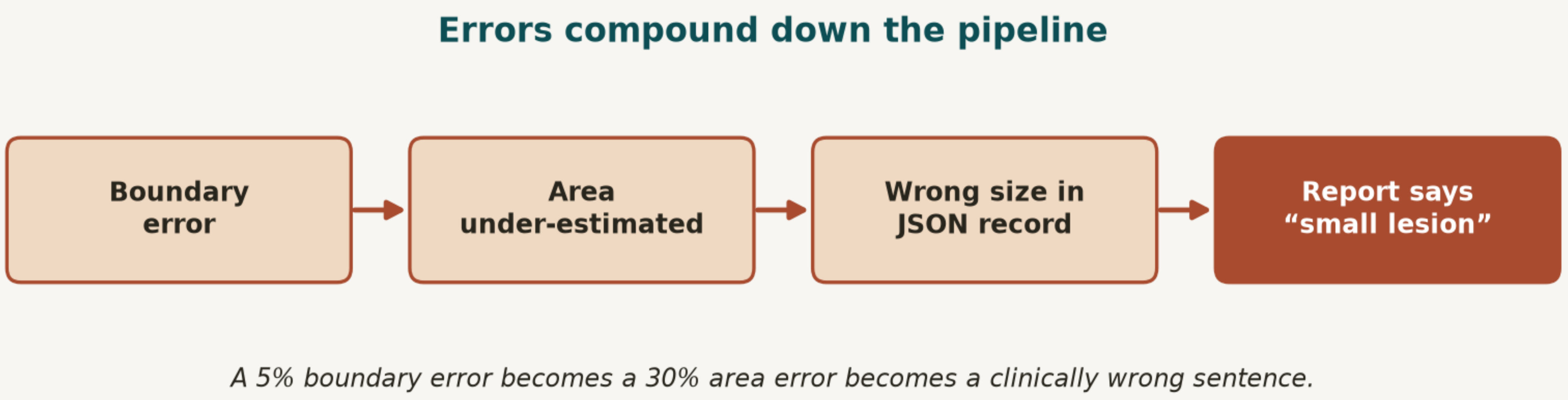
What goes wrong

- Hallucination: the LLM states facts absent from the JSON.
- Bad arithmetic: “two lesions, total area 900” when the record says 1,210.
- Drift: the model is silently swapped; prose style (and errors) change.
- **Overreach: a description turns into a diagnosis or recommendation.**

How you catch it

- Numeric grounding check: every number in the prose must exist in the JSON.
- Role-prompt: “describe only; do not diagnose; say uncertain when unsure.”
- Structured output fields forced before free text.
  - Version the prompt and the model; log both with the report.

Errors compound; they do not cancel



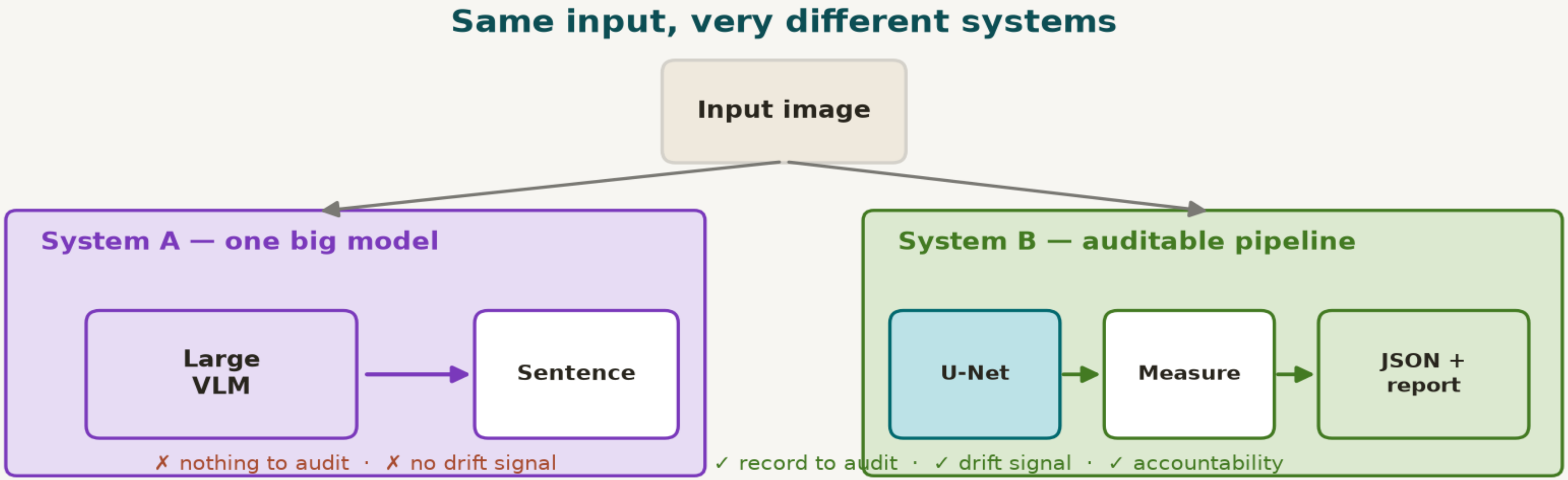
- A small boundary error, multiplied through deterministic measurement and fluent narration, becomes a clinically wrong sentence. Each stage is “almost right”; the chain is wrong.

Push work into the auditable component

Maximise the deterministic, reproducible, structured-record region. Minimise the fluent, irreproducible narrative region.

- ✓ Let deterministic code sum areas and compute a ratio; the LLM only reports the number.
- ✗ Ask the LLM to “estimate the total tumour area from the image”.
- ✓ Apply a threshold rule to flag “unusual”; the LLM reads the flag.
- ✗ Ask the LLM to “decide whether this case is concerning”.

Same inputs, same outputs, very different systems



- Both ship a sentence. Only one can be audited, monitored, repaired, cleared, and held to account.

Never let the narrative invent facts

The narrative may rephrase what is in the structured record. It may not add to it.

The rule, stated

- **Every number in the report must exist in the JSON.**
- Every finding must trace to a measured region.
- Absence in the record = absence in the report.
- "Uncertain" is a valid, encouraged answer.
- **The LLM narrates; it does not perceive, measure, or decide.**

Why it matters

- Hallucinated facts are the defining failure of LLMs in medicine.
- A fluent wrong number is more dangerous than a clunky right one.
- Grounding is enforceable in code; "being careful" is not.
- **This is what makes the system testable: you can assert "report $\subseteq$ record".**

Enforcing grounding through prompt architecture


```
report_prompt.txt

ROLE: You draft findings from a structured record.
RULES:
  1. Use ONLY fields present in <record>.
  2. Copy every number verbatim; do not round.
  3. Do not name findings absent from the record.
  4. Do not diagnose, recommend, or prognose.
  5. If a field is null, write "not assessed".
OUTPUT (JSON then prose):
  {"findings":[], "uncertain":[], "report":"..."}
<record>
  { ...structured JSON from Arrow 3... }
</record>
```

What each rule buys

- **1–3 → grounding (principle 2)**
- **4 → no overreach**
- **5 → honest uncertainty**
- Structured output before prose → machine-checkable.
- <record> injected, not retrieved → no hallucinated context.
  - Prompt + model version logged with every report.

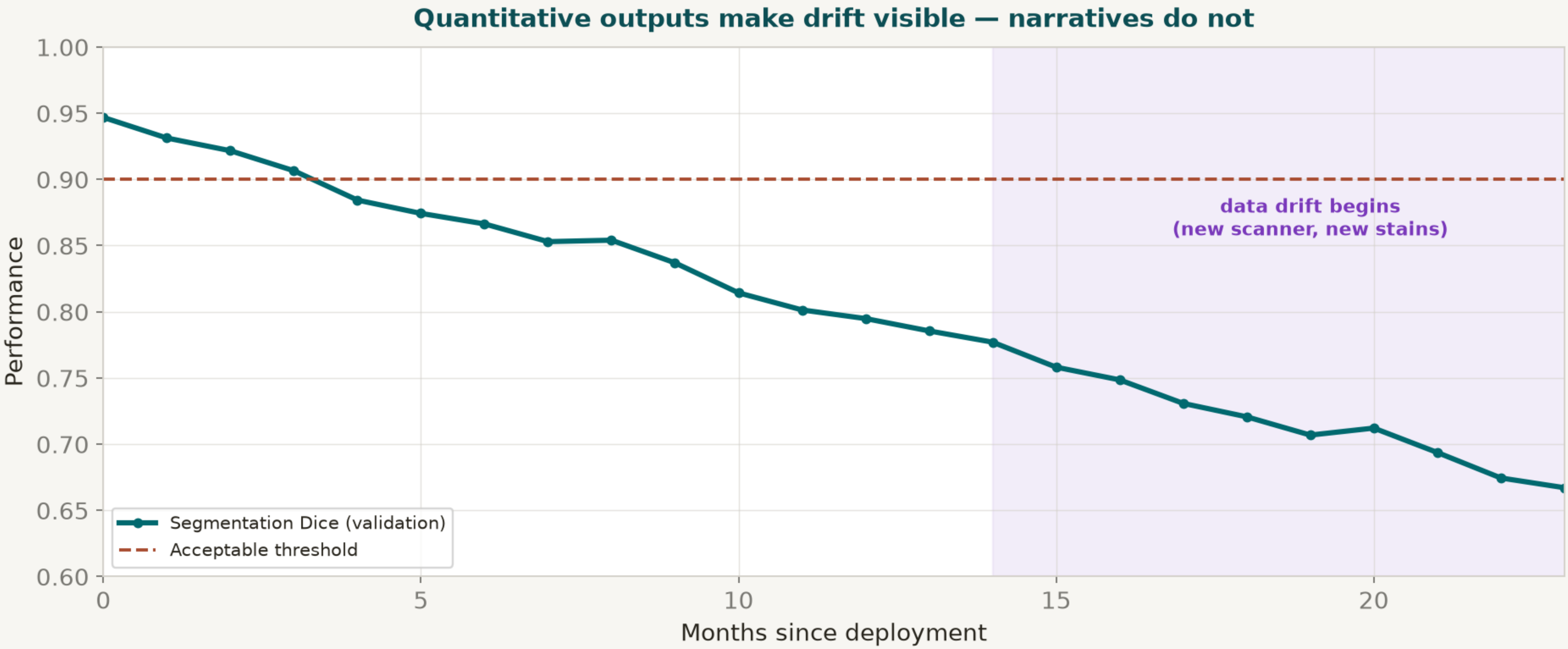
Verifying it: the numeric grounding check

A tiny, deterministic test that runs on every report before a human sees it.

1. Parse the report. Extract every number and every named finding.
2. For each number, assert it exists verbatim in the structured record.
3. For each finding, assert a corresponding measured region exists.
4. For each record field the report omits, decide if silence is acceptable.
5. Fail loudly → route to human queue. Never silently ship an ungrounded report.

"Report $\subseteq$ record" — the one assertion that makes an LLM-in-the-loop system testable.

A pipeline that worked at deployment will not work forever



- New scanner, new stains, new patient population — the distribution shifts. If you log numbers, you see it. If you only log prose, you don't.

Block A in five rules

1. **A medical AI system is a pipeline, not a model.**
2. **A structured, versioned record is the source of truth.**
3. **Push work into the auditable region; starve the narrative.**
4. **The narrative may rephrase the record; it may not add to it.**
5. **Log numbers, so drift is visible before it is fatal.**

BIOMEDICAL IMAGE ANALYSIS · LAB 5 EXTENSION

A hybrid bioimage pipeline, demonstrated on three real datasets

1. **Image** — raw microscopy / photo →
2. **Segmentation** — U-Net · threshold · watershed →
3. **Region properties** — area · shape · intensity →
4. **Text summary** — stats → one sentence →
5. **LLM narration** — JSON record + narrative

✓ Reproducible — temperature 0 + model.eval()

✓ Structured JSON and prose in one call

✓ Audit column catches LLM hallucination

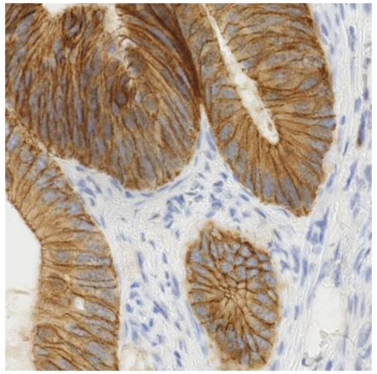
Deterministic measurement you can audit, plus language you can read.

scikit-image segmentation + region properties · Ollama (llama3.2) narration · figures computed on real bundled datasets

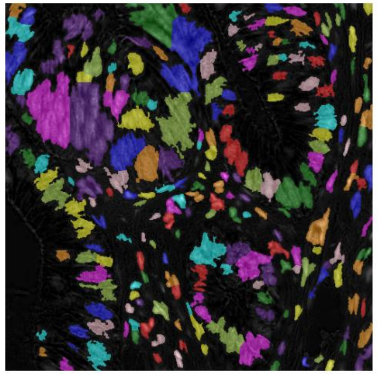
REAL DATASET

Example 1 · Nuclei in stained tissue (IHC)

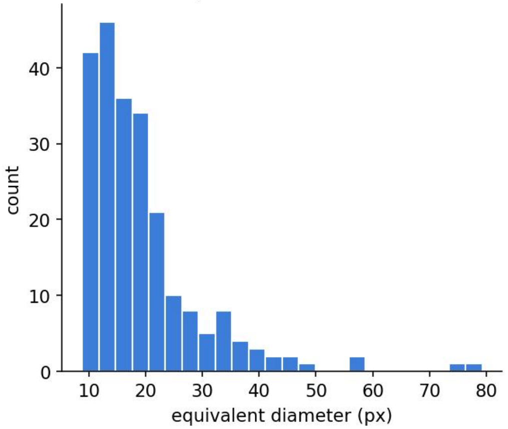
IHC histology (brightfield)
input image



segmentation
226 objects



object size distribution



226 nuclei	**379 px** mean area	**0.79** eccentricity	**0.85** solidity

How *Hematoxylin channel via colour deconvolution → Otsu threshold → watershed split of touching nuclei → regionprops. No labels or training needed.*

STRUCTURED OUTPUT (LLM stage)

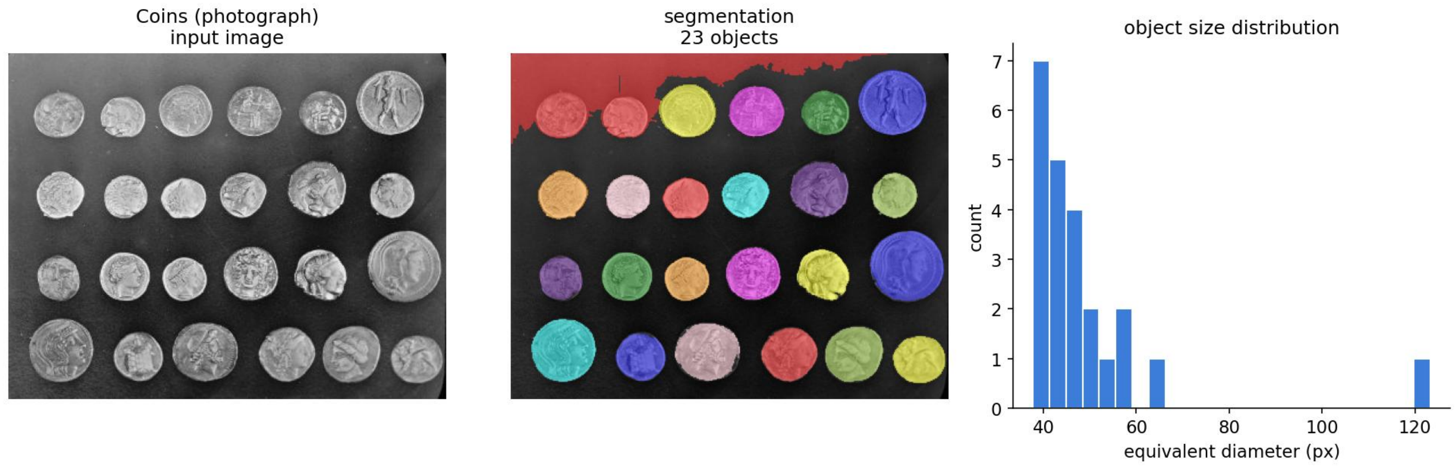

```
{
  "n_objects": 226,
  "density_class": "dense",
  "shape_regularity": "elongated/irregular",
  "size_uniformity": "highly variable",
  "mean_eccentricity": 0.79,
  "mean_solidity": 0.85,
  "quality_flag": "ok"
}
```


NARRATIVE

226 stained nuclei cover 33% of the field — a dense scene. Objects are elongated and only moderately convex (ecc 0.79, solidity 0.85) with highly variable sizes.

REAL DATASET

Example 2 · Coins — same pipeline, no retraining



23	**2119 px**	**0.38**	**0.96**
objects	mean area	eccentricity	solidity

How *Grayscale → Otsu threshold → fill holes → distance-transform watershed to split touching coins → regionprops. Identical code and thresholds as the tissue example.*

STRUCTURED OUTPUT (LLM stage)

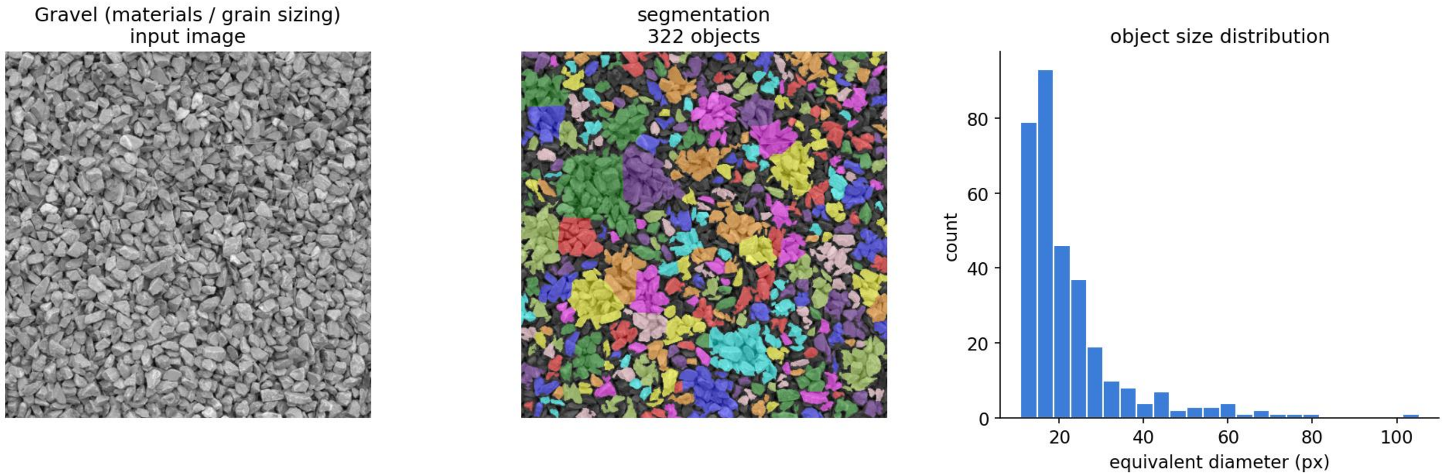

```
{
  "n_objects": 23,
  "density_class": "dense",
  "shape_regularity": "round/regular",
  "size_uniformity": "highly variable",
  "mean_eccentricity": 0.38,
  "mean_solidity": 0.96,
  "quality_flag": "ok"
}
```


NARRATIVE

23 coins, round and highly convex (ecc 0.38, solidity 0.96) — the opposite shape signature to nuclei. Detected true count ≈ 24 with the very same code, no retraining.

REAL DATASET

Example 3 · Gravel — grain sizing in another domain



322	**516 px**	**0.76**	**0.83**
grains	mean area	eccentricity	solidity

How *Grayscale → Otsu threshold → watershed split → regionprops. The size distribution (right panel) is exactly the grain-size histogram a materials workflow needs.*

STRUCTURED OUTPUT (LLM stage)


```
{
  "n_objects": 322,
  "density_class": "dense",
  "shape_regularity": "elongated/irregular",
  "size_uniformity": "highly variable",
  "mean_eccentricity": 0.76,
  "mean_solidity": 0.83,
  "quality_flag": "ok"
}
```


NARRATIVE

322 grains covering 63% of the frame — irregular, less-convex particles (ecc 0.76, solidity 0.83). The equivalent-diameter histogram is a ready-made grain-size distribution.

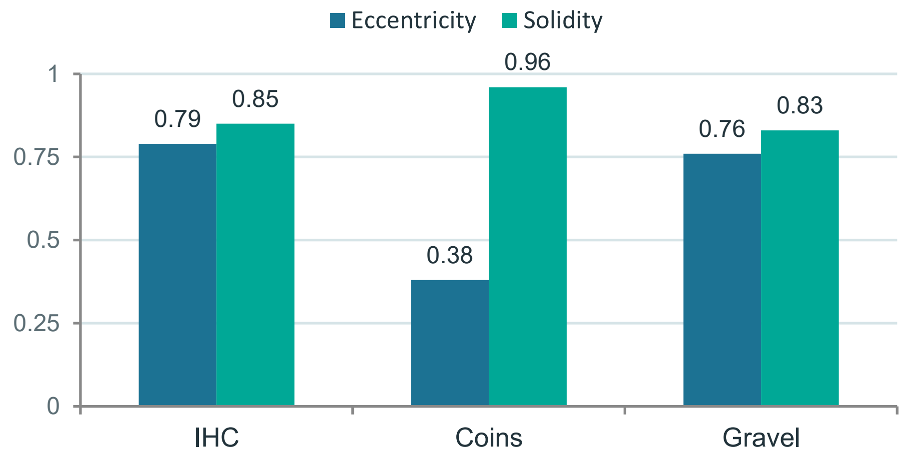
SUMMARY

Results & what this teaches

Dataset	Objects	Mean area (px)	Eccentricity	Solidity	Coverage
IHC histology	226	379	0.79	0.85	33%
Coins	23	2119	0.38	0.96	42%
Gravel	322	516	0.76	0.83	63%

Shape signature: *round + convex coins (ecc 0.38 / sol 0.96) vs irregular nuclei & grains (ecc ~0.76 / sol ~0.84).*

SHAPE SIGNATURE PER DATASET



1. **Domain-agnostic measurement**
   One pipeline counts nuclei, coins, and grains — unchanged.

2. **Language layer adapts**
   The LLM renders the same numbers as prose + JSON per domain.

3. **Auditable by design**
   Measured count rides beside the LLM's count to catch hallucination.

4. **Cheap gate first**
   Deterministic checks reject bad images before a costly LLM call.

BREAK

15 minutes · stretch, then three case studies

BLOCK B

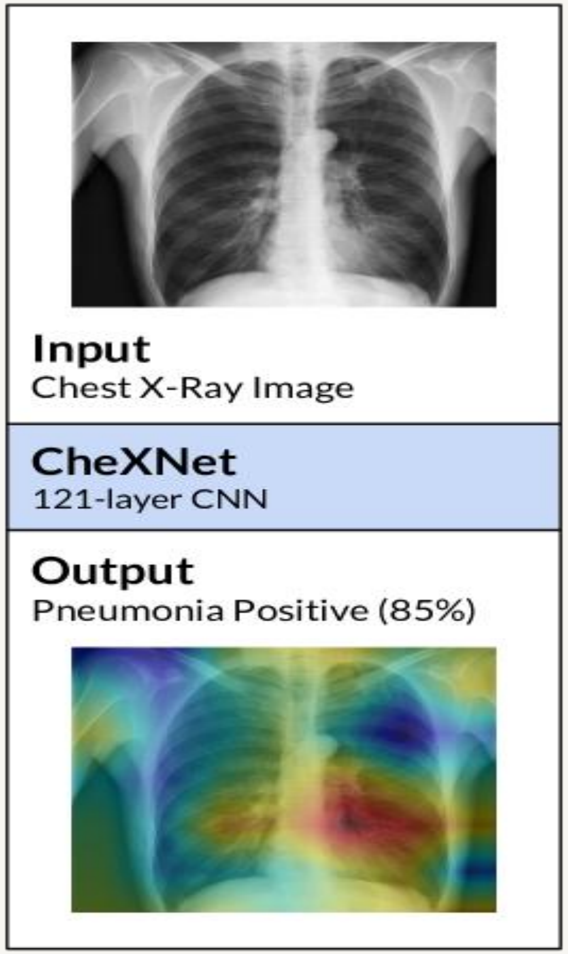
Medical AI in practice

Three case studies · ethics and regulation · where the field is heading

How to read a medical-AI case study

1	**What does each component do?**	Map it to the seven-role anatomy.
2	**Where is the structured record?**	If there isn't one, ask why.
3	**What does it claim — and not claim?**	Read the intended-use statement, not the press release.
4	**How was it evaluated?**	Prospective? Multi-site? Against what ground truth?
5	**Where is the human?**	Who signs the output, and what do they see?

Case 1 — A cleared radiology triage tool



CheXNet: 121-layer CNN, pneumonia detection (Rajpurkar et al. 2017)

The academic precursor

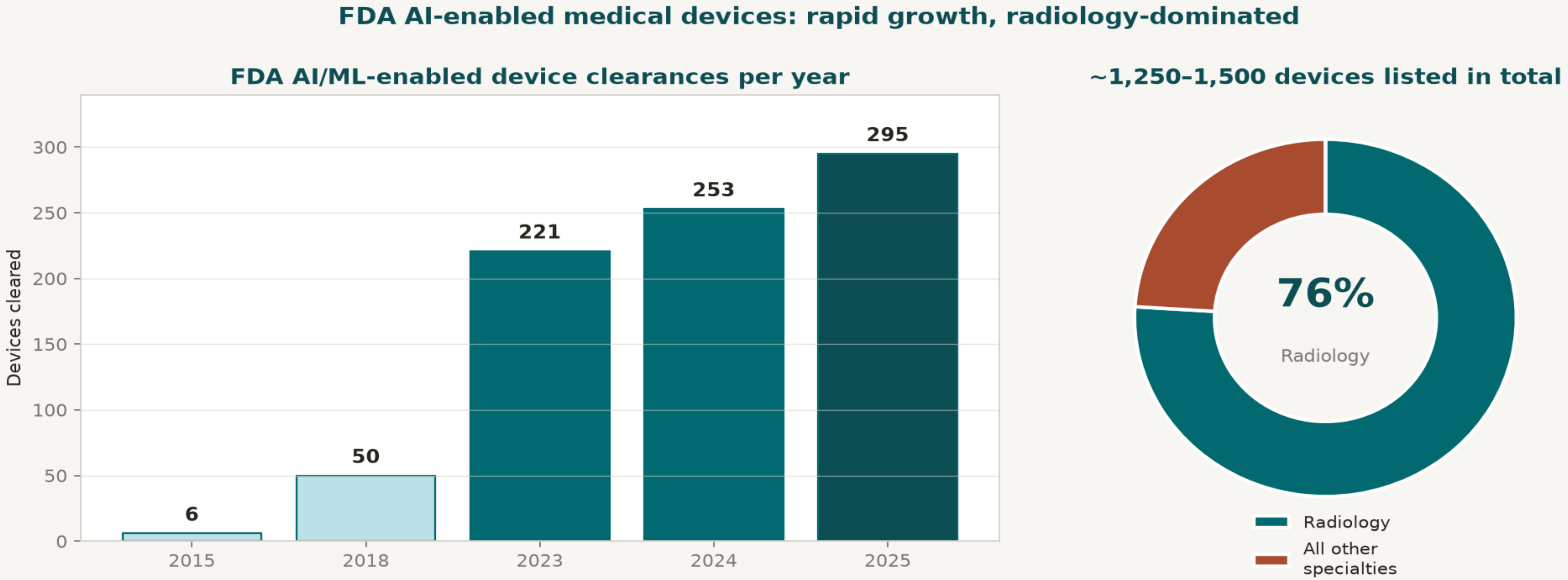
- CheXNet (2017): a 121-layer CNN detects pneumonia on chest X-rays, claiming radiologist-level AUROC.
- **A benchmark triumph. Not a deployed clinical workflow.**
- High benchmark score is not a device you can clear, monitor, or hold to account.

The deployed system

- Aidoc BriefCase and Viz.ai ContaCT: FDA-cleared radiological triage & notification software.
- **They re-order the radiologist's worklist. They do not diagnose.**
  - A cleared device narrows the task; it does not enlarge the model.

Sources: Rajpurkar et al., CheXNet (2017)

FDA AI-enabled medical devices: scale and shape



- Rapid growth, radiology-dominated. Crucially, as of 2025 essentially none use a large language model — every cleared device is a narrow, task-specific model.

Sources: FDA — AI-Enabled Medical Devices

Triage = prioritisation, not diagnosis

Verbatim intended-use language (FDA clearance, Aidoc BriefCase, K213721):

> *"… intended to assist hospital networks … in workflow triage by flagging and communication of suspected positive findings … The device does not alter the original medical image and is not intended to be used as a diagnostic device … assist with triage/prioritization … Notified clinicians are responsible for viewing full images per the standard of care."*

What it claims

- Flag suspected ICH/LVO on images already acquired.
- Move those cases up the radiologist worklist.
- Notify the relevant clinician in parallel.
  - Save minutes-to-hours in stroke triage.

What it does NOT claim

- **It does not diagnose.**
- **It does not exclude disease when negative.**
- It does not read images a human never reads.
  - The clinician still views the full study.

Sources: FDA 510(k) K213721 (Aidoc)

Evaluating a triage tool is not the same as scoring a model

What the paper scores

- AUROC, sensitivity, specificity on a held-out test set.
- Per-class detection of the target finding.
  - Performance under distribution shift.

What the deployment needs

- **Time-to-notification vs. usual workflow (the point of triage).**
- Alert burden & false-positive fatigue on real lists.
- Subgroup sensitivity (scanner, site, age, sex, ethnicity).
- Did outcomes improve, or just reordering? (clinical utility)
  - Viz.ai ContaCT was the first FDA-cleared AI stroke triage (Feb 2018).

Sources: [FDA De Novo DEN170073 (Viz.ai ContaCT)]

Case 2 — A digital pathology research pipeline

Whole-slide images, a segmentation model, and a survival curve. Research-grade — and a preview of how this becomes a clinical assay.

WSI

Whole-slide image acquired, tiled into patches.

Segment

A model (U-Net / foundation model) finds nuclei or regions.

Measure

Deterministic code counts, areas, densities per region.

Record

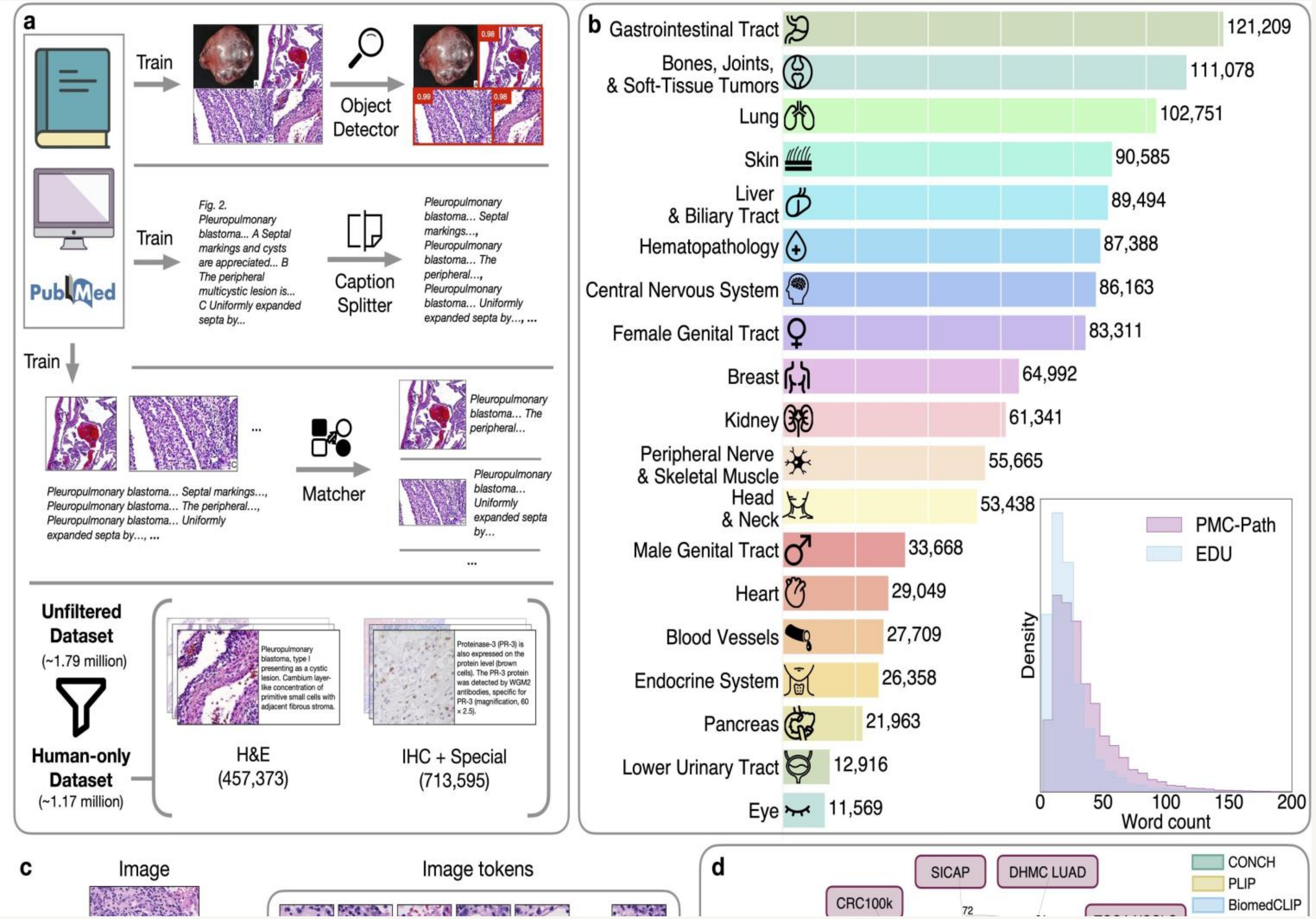
Structured table: one row per slide, features + outcome.

Analyse

Statistics links features to survival — not the model.

- The pipeline is identical in shape to Lab 5: image → mask → measurement → structured record → analysis. Only the modality (a gigapixel H&E slide) and the downstream question (does this feature predict survival?) differ.

CONCH: a pathology foundation model



CONCH: vision-language pretraining on ~1.17M pathology image–caption pairs (Lu et al., Nat. Med. 2024)

Why it matters here

- **A foundation model for pathology: one backbone, many downstream tasks.**
- Image–caption pairs from PubMed textbooks → broad tissue coverage.
- Zero-shot classification, retrieval, segmentation downstream.
- **It is one component (the Segment box) in a pipeline — not the pipeline.**

Sources: [Lu et al., CONCH (Nat. Med. 2024)]

How segmentation becomes a prognosis

Segment	tumour / stroma / lymphocyte regions per slide
Measure	area fractions, counts, density per region
Reduce	one feature vector per patient (the structured record)
Correlate	feature vs. survival — a Cox model, not a neural net
Interpret	“high lymphocyte density predicts longer survival”

- The model finds regions. Statistics, on the structured record, finds the association. The survival curve is a consequence of the pipeline — not an output of the network.

Same shape, new failure modes

Acquisition & slide-level

- Stain variation across labs: H&E is not standardised.
- Scanner differences: colour, focus, tile alignment.
- Tissue folds, artefacts, out-of-focus regions.
- **Tile-level stats ≠ slide-level truth (aggregation errors).**

Model & statistics

- Segmentation trained on one organ fails on another.
- Survival association confounded by site / preprocessing.
- Feature selected on training set → overfit, no hold-out.
- **A prognostic biomarker needs its own validation, not just AUROC.**

Case 3 — An LLM radiology report drafter

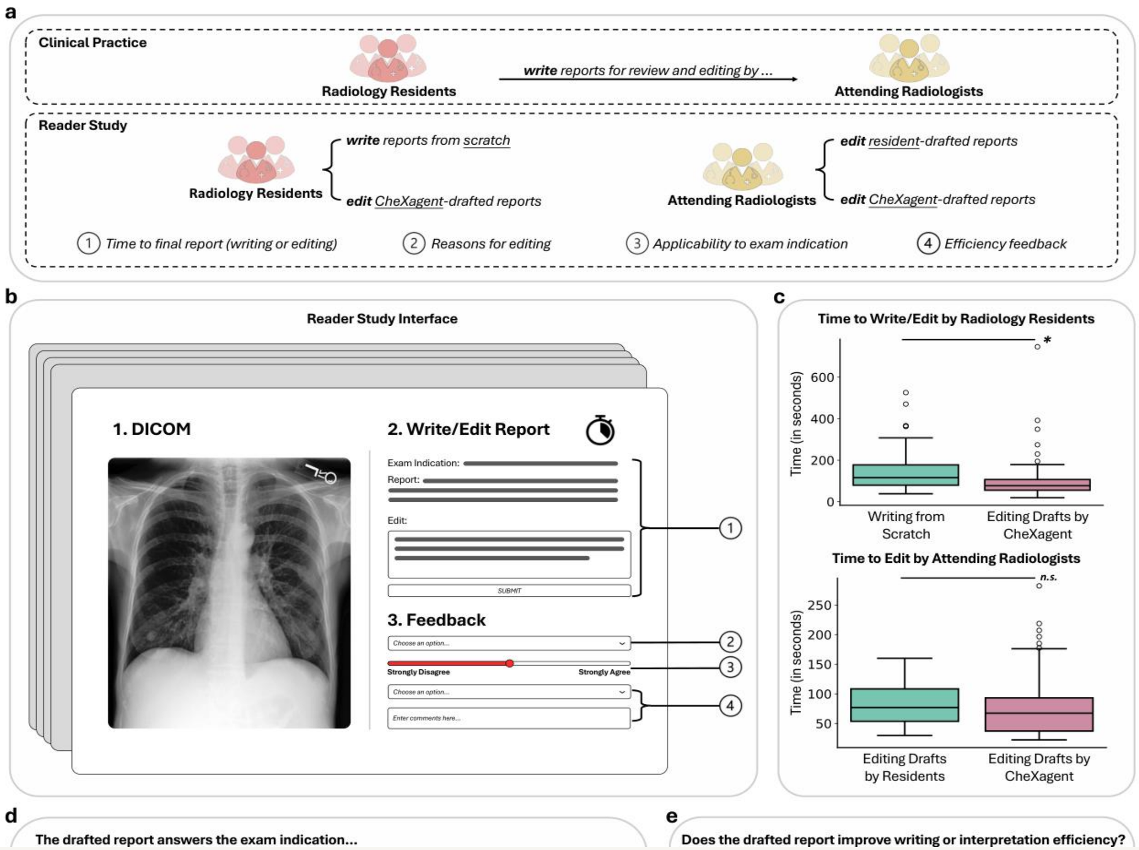
A language model drafts a report from the structured findings. A radiologist edits and signs it. Human-in-the-loop, auditable, grounded.

Findings (structured) → LLM drafts report → Radiologist reviews & edits → Signed final report

What makes this a pipeline, not a model

- Drafts are grounded in the structured findings; numbers are checked; the human signs. The LLM accelerates; it does not replace. Drift is visible because editing effort & grounding-pass-rate are logged.

CheXagent: human-in-the-loop report drafting



CheXagent reader study: clinical practice vs. editing model drafts (Chen et al. 2024, Fig. 6)

The headline result

- **Editing CheXagent drafts cut resident report-writing time by ~36%.**
- 156.4s → 99.9s per study.
  - 8-radiologist prospective reader study.
- **Drafts are edited and signed by a human — triage of attention, not diagnosis.**

Sources: Chen et al., CheXagent (2024)

The human is not a fallback — they are the accountability

Design for review

- Show the structured findings beside the draft, not just prose.
- Highlight low-confidence / low-grounding sections for the editor.
- Make edits feed back: track what humans change (a free audit).
- **“Accept as-is” should require an active action.**

Failure modes of human review

- Automation bias: humans accept plausible-sounding drafts.
- Time pressure erodes the check the system was built for.
- No metric on review depth → review quietly degrades.
  - Log edit distance & override rate; investigate drift.

Scoring a draft is not scoring a workflow

Time

Editing time vs. writing from scratch (the productivity claim).

Accuracy

Final signed report vs. consensus ground truth — not the draft.

Grounding

% of report numbers present in the record (the pipeline invariant).

Edits

Edit distance draft→final; what humans keep changing reveals blind spots.

Subgroups

Performance across site, scanner, age, sex, ethnicity.

Safety

Missed findings, added findings (hallucination rate) per thousand reports.

ETHICS & REGULATION

Research use is not clinical use.

Bias in, harm out. Clearance is a floor, not a ceiling.

Research use is not clinical use

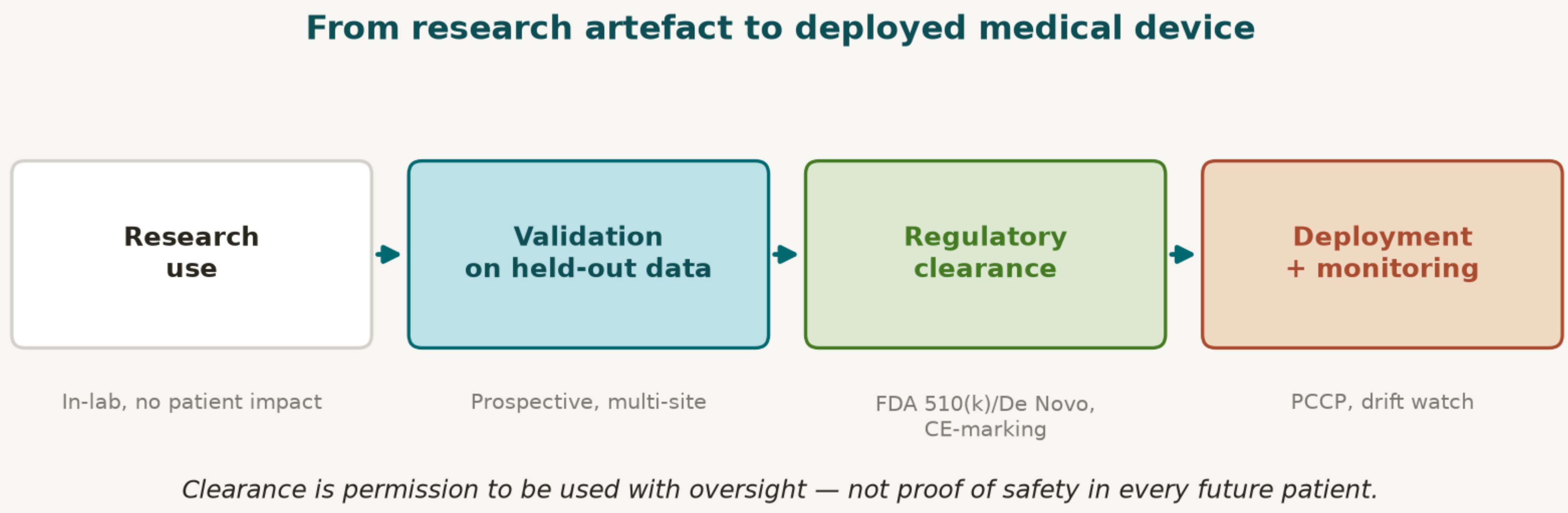
Research use

- A model that beats a benchmark on a public dataset.
- No obligation to patients; no intended-use contract.
- Failure is a paper revision.
  - “Works” = statistically better on the test set.

Clinical use

- A device with a cleared intended use and a named manufacturer.
- Obligation to patients; post-market surveillance.
- **Failure is harm — and liability.**
  - “Works” = improves outcomes, safely, in your population.

The regulatory pathway, simplified



- Clearance authorises a specific intended use. The Predetermined Change Control Plan (PCCP, FDA guidance finalised Dec 2024) pre-authorises specified model updates — so monitoring can actually lead to change.

Sources: FDA — PCCP guidance (2024)

What “FDA-cleared / CE-marked” actually means

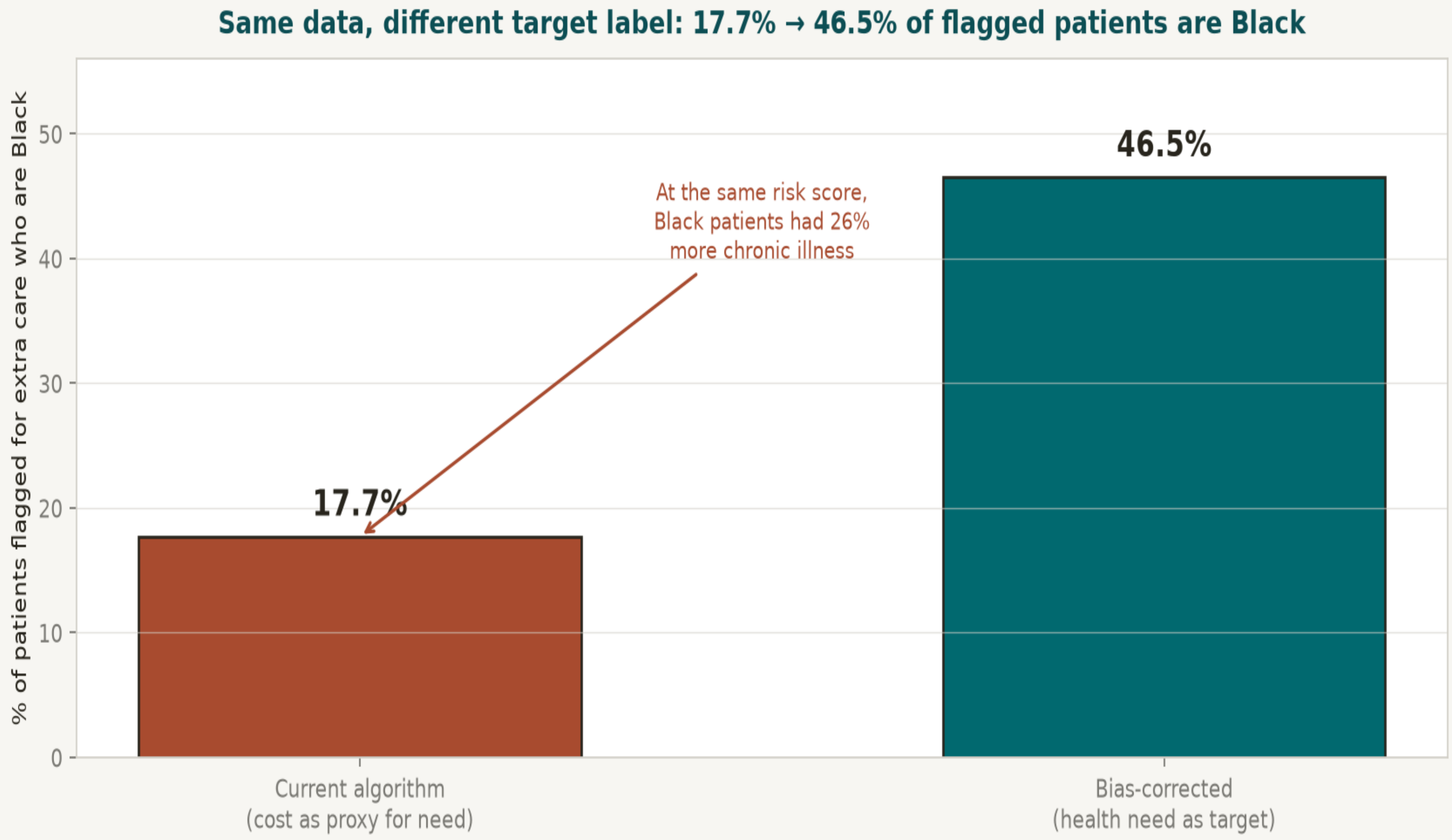
It means

- The device is safe & effective for a specific, stated intended use.
- A named manufacturer is accountable.
- Performance was demonstrated on representative data.
  - Post-market obligations exist (reporting, monitoring).

It does NOT mean

- **It works in your population, on your scanner, for your task.**
- **It is unbiased.**
- It is autonomous — most require the clinician in the loop.
- It will keep working as the world changes.

Bias in training data becomes harm in deployment



The Obermeyer finding

- **A widely-deployed US health-risk algorithm used healthcare cost as a proxy for need.**
- Because Black patients had less spent on them at equal need, the algorithm under-flagged them.
- **A bias-free version would flag 46.5% Black patients vs 17.7% today.**
    - Fixing the target label reduced bias 84% — it was a design choice.

Sources: Obermeyer et al., Science (2019)

What an average conceals

A model with good overall accuracy can still under-serve the patients who need it most.

The finding (Seyyed-Kalantari 2021)

- Three chest X-ray datasets: MIMIC-CXR, CheXpert, ChestX-ray14.
- **Models underdiagnosed: female, Black, Hispanic, young, Medicaid patients.**
- Overall AUROC looked fine — the gap was hidden in subgroups.
- **Bias was invisible without subgroup reporting.**

The pipeline lesson

- **Log every prediction with subgroup metadata.**
- Report subgroup metrics, not just the headline.
- Bias is a property of the data + the task, not the architecture.
  - A structured record makes subgroup audit possible.

Sources: Seyyed-Kalantari et al., Nat. Med. (2021)

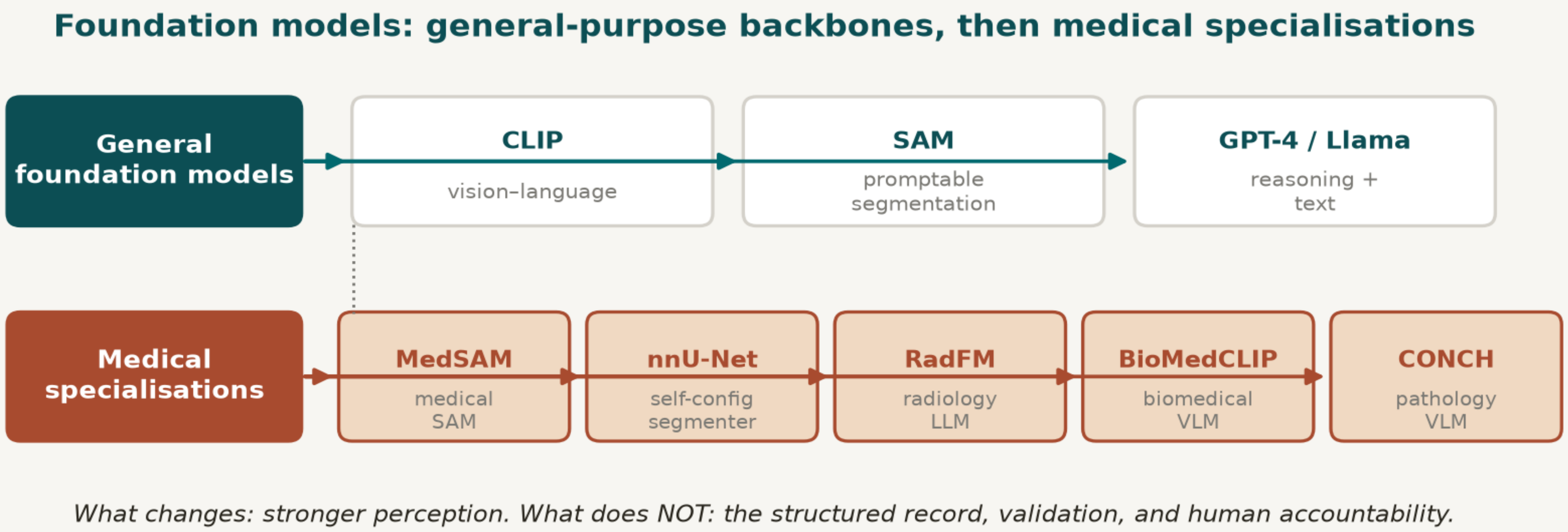
The accountability gap

When a pipeline fails and harms a patient, who is responsible?

Model authors	*"We published a method; deployment wasn't us."*
Hospital IT	*"We installed what was cleared; we don't train models."*
Clinician	*"I trusted the system; the case looked normal."*
Manufacturer	*"It was used outside its intended use."*
Patient	*(harmed, and not at the table)*

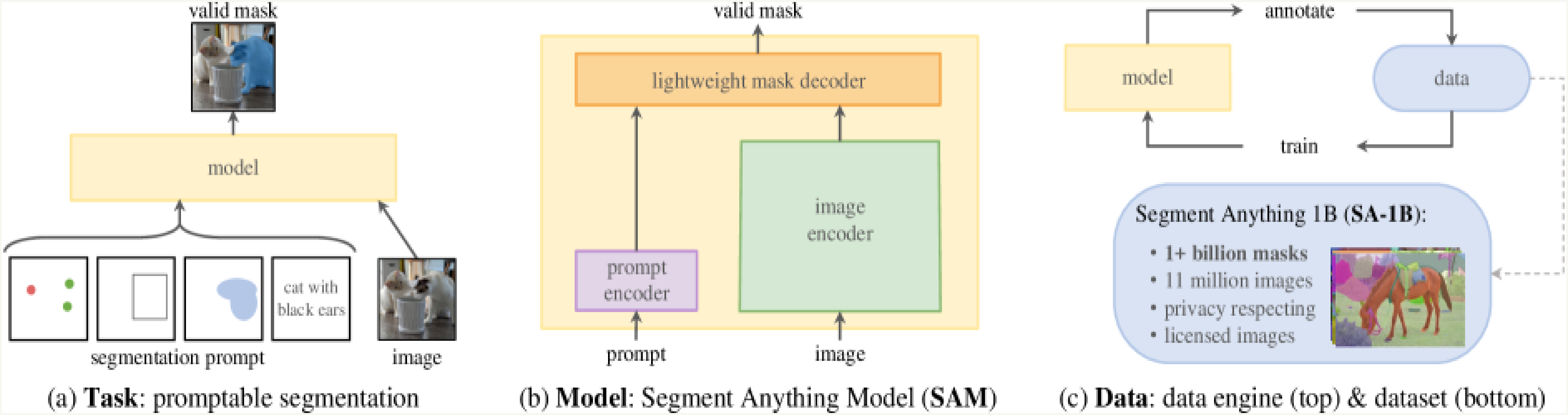
A structured record with named owners at each step is the technical answer to a social problem.

Foundation models: what changes



- One large, pre-trained backbone (SAM → MedSAM → RadFM / CONCH / BioMedCLIP) replaces the hand-built segmentation or classification box — but only that box.

Segment Anything (SAM)

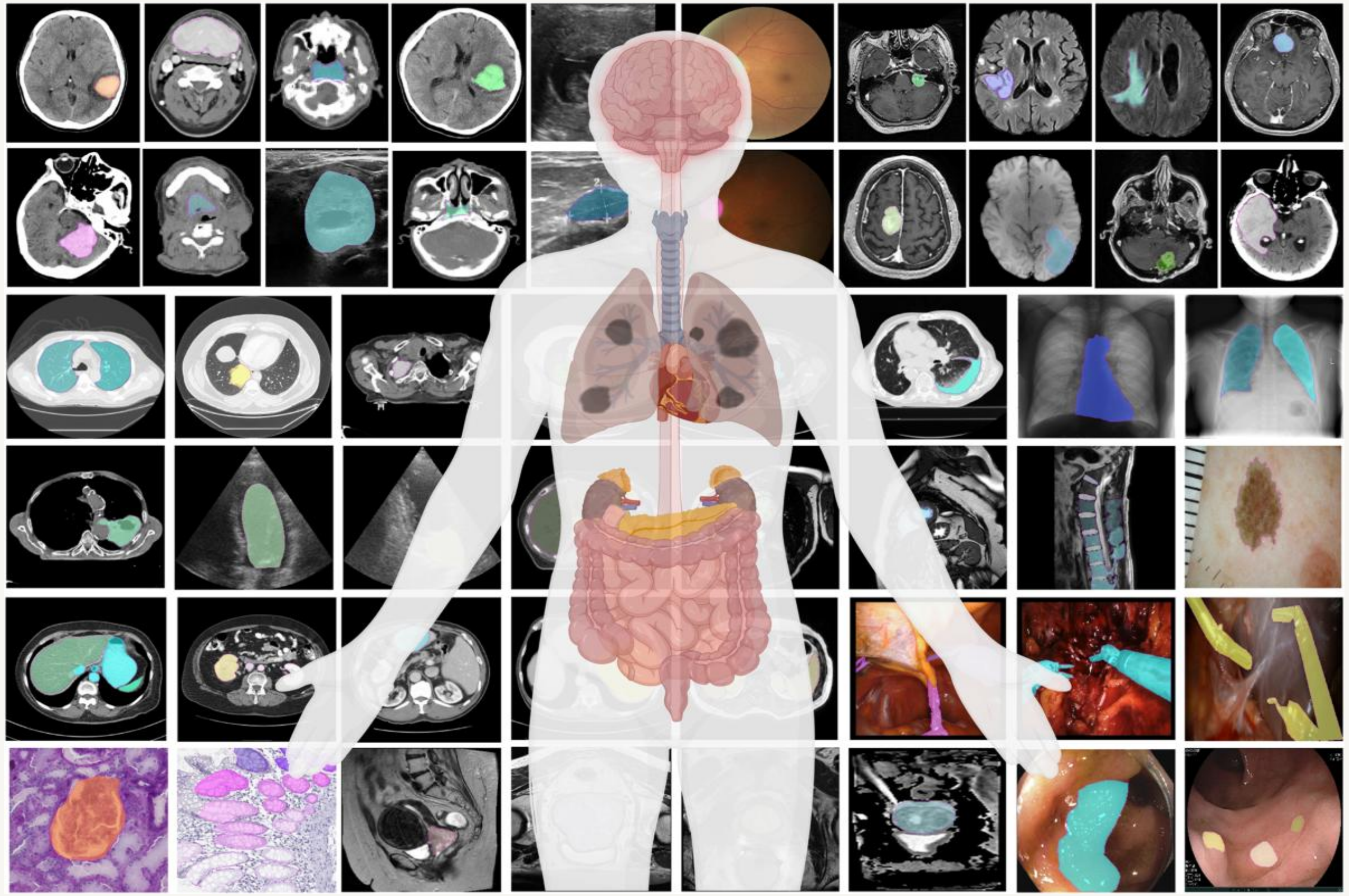


SAM: promptable segmentation via image encoder, prompt encoder, mask decoder (Kirillov et al., ICCV 2023)

- A general-purpose, promptable segmenter trained on 1B masks. A building block — when you point at an object, it segments it. It is one component; it still needs measurement, a record, and a human.

Sources: Kirillov et al., Segment Anything (2023)

MedSAM: adapting SAM to medical imaging



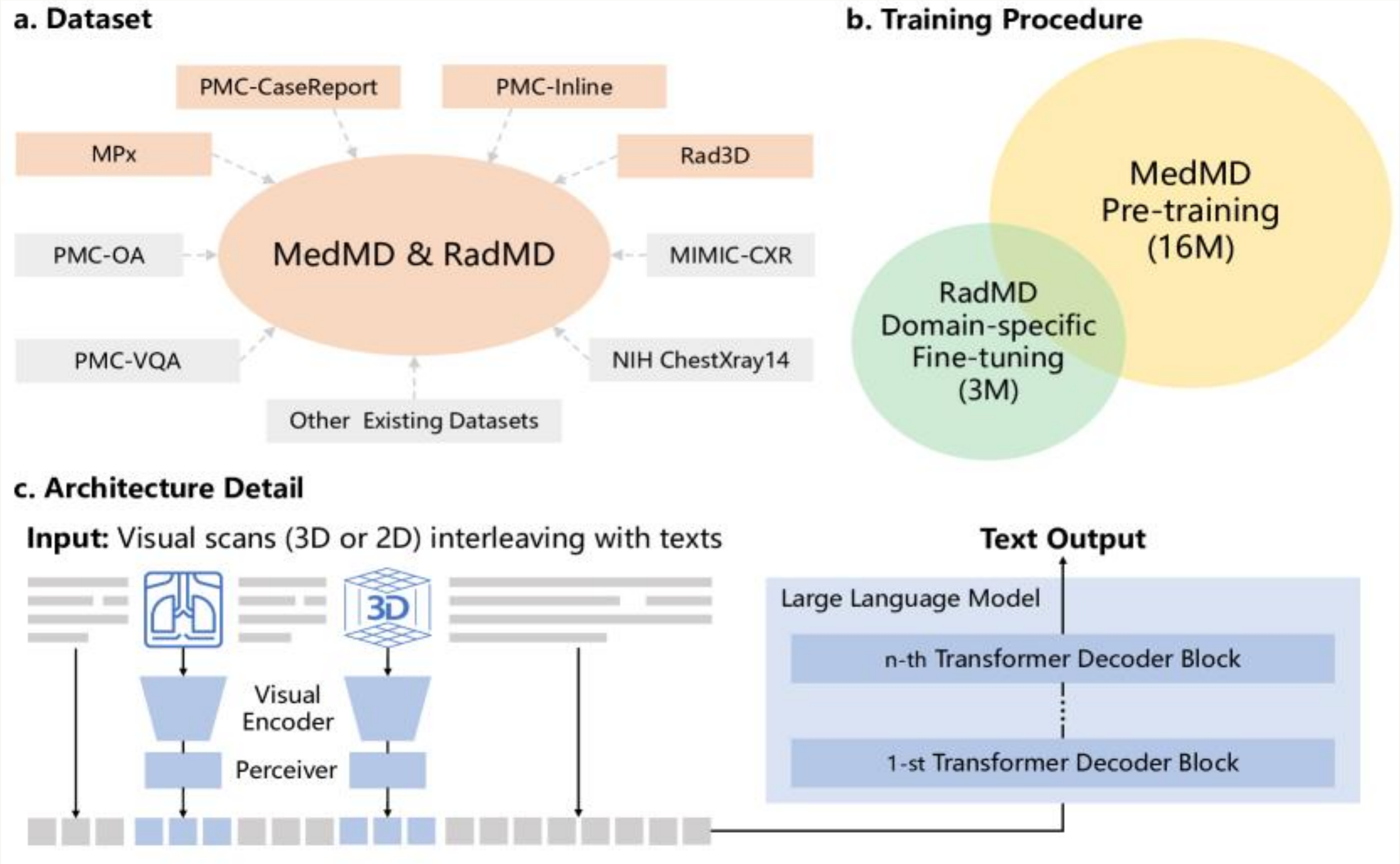
MedSAM: promptable segmentation across CT, MRI, X-ray, ultrasound, pathology (Ma et al., Nat. Commun. 2024)

What it teaches

- **A general foundation model, fine-tuned on >1M medical mask–prompt pairs.**
- One model, many modalities & organs.
- **Still a component: it produces masks, not diagnoses.**
  - It must sit inside the same auditable pipeline.

Sources: Ma et al., MedSAM (Nat. Commun. 2024)

RadFM: a radiology foundation model



RadFM: visual encoders + Perceiver + LLM for radiology (Wu et al. 2023)

What it teaches

- **Attempts to put perception + language in one model.**
- **Impressive on benchmarks; not a deployed, cleared device.**
- The “one big model” temptation, on full display.
  - See slide 6: you give up testability, repair, and accountability.

Sources: [Wu et al., RadFM (2023)]

Foundation models: what does NOT change

A better component does not retire the pipeline.

- ✓ **The structured record is still the source of truth.**
- ✓ **Measurement is still deterministic and owned.**
- ✓ **The narrative is still grounded, checked, and signed.**
- ✓ **Validation is still per-population, per-subgroup.**
- ✓ **Monitoring and drift detection still run on logged numbers.**
- ✓ **A named human is still accountable for the output.**

Swapping the model changes the third box. It changes nothing on this list.

Which step would you trust least?

Take five minutes in pairs. Pick one of the three case studies and answer:

1. Which arrow in its pipeline would you trust least, and why?
2. If that arrow failed silently, how long before anyone noticed?
3. What single logged quantity would let you catch it?
4. Who, by name, is accountable if it fails — and are they at the table?

Reading list

U-Net (2015)	The segmentation component.	link
nnU-Net (2021)	A pipeline, not a model.	link
SAM (2023) & MedSAM (2024)	Foundation segmentation models.	link
CONCH (2024)	Pathology foundation model.	link
CheXagent (2024)	Human-in-the-loop report drafting.	link
Obermeyer (2019) & Seyyed-Kalantari (2021)	Algorithmic bias in deployment.	link
FDA AI-Enabled Medical Devices	The cleared-device landscape.	link

TAKE-AWAY

A medical AI system is a pipeline with a structured record, a grounded narrative, and a named human who signs it.

Design the pipeline. Maximise the auditable part. Forbid the narrative from inventing facts. Log the numbers. Read the intended-use statement. And remember which step you would trust least.